

16 Discrete Random Variables: Probability Mass Functions

- Roughly, a random variable assigns a number measuring some quantity of interest to each outcome of a random phenomenon.
- The **(probability) distribution** of a random variable specifies the possible values of the random variable and a way of determining corresponding probabilities.
- We will see many ways of describing a distribution, depending on how many random variables are involved and their types (discrete or continuous)
- A **discrete** random variable can take on only countably many isolated points on a number line. These are often counting type variables.
 - Note that “countably many” includes the case of countably infinite, such as $\{0, 1, 2, \dots\}$.
- The **probability mass function (pmf)** of a *discrete* RV X , defined on a probability space with probability measure P , is a function $p_X : \mathbb{R} \mapsto [0, 1]$ which specifies each possible value of the RV and the probability that the RV takes that particular value: $p_X(x) = P(X = x)$ for each possible value of x .
- Certain common distributions have special names and properties. We often specify a distribution by name along with values of relevant parameters.

Example 16.1. Roll a fair four-sided die twice and let X be the sum and Y the larger of the two rolls.

1. Verify that the pmf of X is

$$p_X(x) = \frac{4 - |x - 5|}{16}, \quad x = 2, 3, \dots, 8$$

2. Is the following the pmf of Y ? Discuss.

$$p_Y(u) = \frac{2u-1}{16}$$

Example 16.2. Randomly select a county in the U.S. Let X be the leading digit in the county's population.

For example, if the county's population is 10,040,000 (Los Angeles County) then $X = 1$; if 3,170,000 (Orange County) then $X = 3$; if 283,000 (SLO County) then $X = 2$; if 30,600 (Lassen County) then $X = 3$. The possible values of X are $1, 2, \dots, 9$. You might think that X is equally likely to be any of its possible values. However, a more appropriate model is to assume that X has pmf

$$p_X(x) = \begin{cases} \log_{10}(1 + \frac{1}{x}), & x = 1, 2, \dots, 9, \\ 0, & \text{otherwise} \end{cases}$$

This distribution¹ is known as [Benford's law](#).

1. Compute and interpret $P(X = 1)$.
2. Compute and interpret $P(X = 2)$.
3. Construct a table specifying the distribution of X , and the corresponding spinner.

¹You can find a nice explanation of Benford's law [here](#).

4. Verify that p_X is a valid pmf.

5. Find $P(X \geq 3)$

- In a sequence of **Bernoulli(p) trials**

- There are only two possible outcomes, “success” (1) or not (0, “failure”), on each trial.
- The unconditional/marginal probability of success is the same on every trial, and equal to p .
- The trials are independent.

Example 16.3. Maya is a basketball player who makes 40% of her three point field goal attempts. Suppose that she attempts three pointers until she makes one and then stops. Let X be the total number of shots she attempts. Assume shot attempts are independent.

1. Can Maya’s attempts be consider Bernoulli trials?

2. What are the possible values that X can take? Is X discrete or continuous?

3. Compute and interpret $P(X = 1)$.

4. Compute and interpret $P(X = 2)$.
5. Compute and interpret $P(X = 3)$.
6. Find the probability mass function of X .
7. Construct a table, plot, and spinner representing the distribution of X .
8. Compute and interpret $P(X > 3)$. Can you think of a way to do this without summing several terms?
9. Compute and interpret $P(X > 5 | X > 2)$. Compare to the previous part. What do you notice? Does this make sense?
10. What seems like a reasonable shortcut formula for $E(X)$ in terms of p ? Hint: it might help to consider the case $p = 0.1$ first, then apply similar logic to $p = 0.4$ and general p .

11. Compute $E(X)$. Did the shortcut work?

12. Interpret $E(X)$ in context.

13. Compute $\text{Var}(X)$.

14. Would $\text{Var}(X)$ be bigger or smaller if $p = 0.9$? If $p = 0.1$?

- Perform Bernoulli(p) trials until a success occurs and then stop. Let X count the total number of trials, including the single success. The distribution of X is defined to be the **Geometric(p) distribution**.
- The Geometric(p) probability mass function follows from the fact that if X has a Geometric distribution, then $X = x$ if and only if
 - the first $x - 1$ trials are failures, and
 - the x th (last) trial results in success.
- A discrete random variable X has a Geometric(p) distribution if and only if its probability mass function is

$$p_X(x) = p(1 - p)^{x-1}, \quad x = 1, 2, 3, \dots$$

- If X has a Geometric(p) distribution then

$$P(X > x) = (1 - p)^x, \quad x = 1, 2, 3, \dots$$

$$E(X) = \frac{1}{p}$$

$$\text{Var}(X) = \frac{1 - p}{p^2}$$

Example 16.4. Continuing the previous example.

1. Specify how you could simulate a value of X using the “simulate from the probability space” method.
 2. Specify how you could simulate a value of X using the “simulate from the distribution” method.
 3. Explain how you could use simulation to approximate the distribution of X and its expected value.
- There are two basic methods for simulating a value x of a random variable X .
 - **Simulate from the probability space.** Simulate an outcome ω from the underlying probability space and set $x = X(\omega)$.
 - **Simulate from the distribution.** Simulate a value x directly from the distribution of X .
 - The “simulate from the distribution” method corresponds to constructing a spinner representing the marginal distribution of X and spinning it once to generate x . This method does require that the distribution of X is known. However, as we will see in many examples, *it is common to specify the distribution of a random variable directly without defining the underlying probability space.*

Example 16.5. Donny Dont is thoroughly confused about the distinction between a random variable and its distribution. Help him understand by providing a simple concrete example of *two different random variables* X and Y that have the *same distribution*. Can you think of X and Y for which $P(X = Y) = 0$?

- Do not confuse a random variable with its distribution!
- A random variable measures a numerical quantity which depends on the outcome of a random phenomenon
- The distribution of a random variable specifies the long run pattern of variation of values of the random variable over many repetitions of the underlying random phenomenon.